

TASK 3 REPORT – WEB APPLICATION SECURITY

Title:

Web Application Security Testing Using DVWA

Objective:

To identify and understand common web vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), CSRF, File Inclusion, and request manipulation using a controlled lab environment.

Tools Used:

- Kali Linux
- DVWA (Damn Vulnerable Web Application)
- Burp Suite
- Web Browser
- Apache / PHP / MySQL

Methodology:

1. Installed and configured DVWA in Kali Linux.
2. Set DVWA security level to low for testing.
3. Tested multiple OWASP vulnerabilities.
4. Captured screenshots of successful exploitation.
5. Documented prevention methods.

Activities Performed:

1. SQL Injection

Entered malicious input in login/search fields to bypass validation and extract data.

2. Cross-Site Scripting (XSS)

Tested reflected and stored XSS using script payloads.

3. Cross-Site Request Forgery (CSRF)

Observed how unauthorized actions can be triggered through crafted requests.

4. File Inclusion

Tested inclusion of local files to access sensitive system data.

5. Burp Suite Testing

Intercepted and modified HTTP requests for analysis.

Findings:

- Input validation weaknesses were present.

- Unsanitized parameters allowed script execution.
- Weak request validation enabled CSRF risk.
- Improper file handling could expose sensitive files.

Mitigation:

- Use prepared statements.
- Validate and sanitize input.
- Use CSRF tokens.
- Implement Content Security Policy (CSP).
- Restrict file access and use allowlists.
- Secure session management.

Conclusion:

This task demonstrated practical web security testing in DVWA and improved understanding of common web application vulnerabilities and their prevention.

Result:

Web vulnerabilities were successfully tested and documented in a safe lab environment.